Lesson 1: What is 3D Printing?

3D Printing Center - Village School

v1

2025

Welcome to 3D Printing!

Establish Class Expectations

What are our 3D Printing M&Ms?

Establish Class Expectations

- Model Respect
  - respect centermates' work
  - respect the printers
    - don't get hurt by them
    - don't hurt them
  - respect the mere 45 minutes per center lesson
- Make Good Choices
- Solve Problems
  - If you know how, help your peers

Welcome to 3D Printing!

Today's Big Question

What is 3D printing and how does it work?

How Does 3D Printing Work?

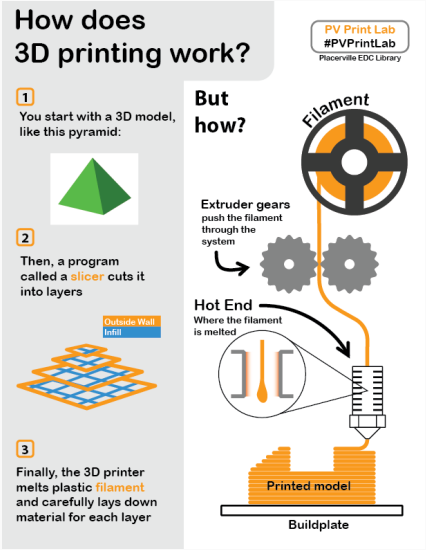
Think of Building with Pancakes!

- 3D printing builds objects **layer by layer**
- Like stacking pancakes on top of each other
- Each layer sticks to the one below it

From Computer to Object

The 3 Steps

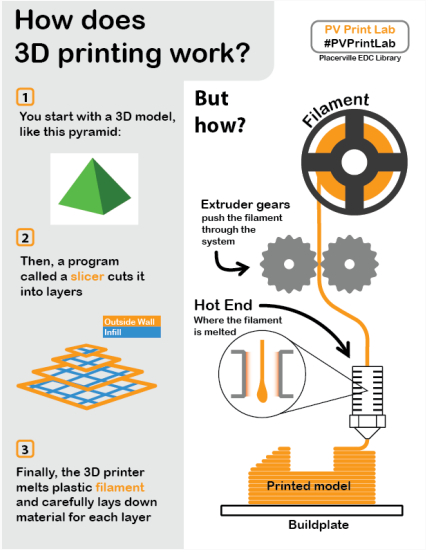
1. **Design** - We create on computers
2. **Slice** - Computer breaks design into layers
3. **Print** - Machine builds layer by layer



The Magic Material

How Plastic Becomes Objects

- Plastic filament gets **heated up**
- Squeezed out like **frosting from a tube**
- **Cools down** and hardens
- Layers **stick together** as they cool



Safety First!

3D Printer Safety Rules

- **Never touch** the hot parts
- **Don't reach** into the printer while it's working
- **Tell a teacher** if something looks wrong
- The printer gets **very hot** - like an oven!

Amazing 3D Printed Objects

Look at These!

- Toys and games
- Tools and gadgets
- Decorations
- Food (yes, food!)

Pass around demonstration objects

Real-World Examples

3D Printing Helps People

- **Custom phone cases** and game pieces
- **Replacement parts** for broken toys
- **Art projects** and sculptures
- **Helpful tools**
- **Medical devices** like prosthetics
- **Aerospace** and car parts

Interactive Activity

Object Detective

Examine the printed objects on your table

- What do you think each object is used for?
- Can you see the layers?
- Which one is your favorite? Why?

Live Demonstration

Watch the Printer Work!

Show the actual 3D printer in action

- **Hot end** - where plastic comes out
- **Build plate** - where object grows
- **Filament** - the plastic "thread"

What We Learned Today

3D Printing is Amazing!

- Builds objects **layer by layer**
- Starts with **computer designs**
- Uses **heated plastic**
- Helps people in **many ways**
- Is used in **many industries**

Coming Up Next

Lesson 2 Preview

“Now that we know what 3D printing is, we’ll learn to design our own objects!”

- Learn to use Tinkercad
- Design your first keychain
- See your ideas come to life!

Reflection Time

What surprised you most about 3D printing?

Quick sharing - one word or phrase

Cleanup Time!

5 Minutes to Pack Up

- Help organize demonstration materials
- Put objects back carefully
- Clean up your workspace
- Get ready for next class

Great job learning about 3D printing today!